PROVISIONAL APPLICATION FOR PATENT

Title of the Invention

ATOMIK: An Externally Stateless, Delta-Driven Computing Architecture for Deterministic State Evolution

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Cross-Reference to Related Applications

None.

Federally Sponsored Research

Not applicable.

Field of the Invention

The present invention relates to computer architecture, data processing systems, and computational state management. More specifically, the invention relates to externally stateless computing architectures that encode computation as deterministic state transitions using register-local logic and delta representations.

Background of the Invention

Conventional computing architectures rely on persistent memory systems in which data is repeatedly stored, retrieved, modified, and rewritten. This store–process–store paradigm introduces latency, bandwidth bottlenecks, and energy inefficiencies, particularly as processor speeds outpace memory access performance, a limitation commonly referred to as the memory wall.

Persistent state storage also introduces security vulnerabilities, including susceptibility to memory scraping, side-channel attacks, and unauthorized data exfiltration. These limitations are amplified in distributed, edge, embedded, and real-time systems.

Accordingly, there exists a need for a computing architecture that decouples computational throughput from persistent memory access, minimizes data movement, and reduces security risks associated with data at rest.

Summary of the Invention

The present invention, referred to as ATOMIK, provides an externally stateless computing architecture in which computation is represented as deterministic state evolution rather than persistent state storage. The architecture operates by computing deltas between successive data representations using register-local logic, thereby encoding only changes in system state.

Incoming data atoms are mapped to registers and compared against prior state values using a delta computation operation. The resulting delta stream fully represents system evolution without requiring storage of raw data or persistent external memory.

Advantages include reduced bandwidth consumption, deterministic low-latency execution, improved energy efficiency through minimized data movement, and enhanced security due to the absence of durable data at rest.

Detailed Description of the Invention

Externally Stateless Architecture

The architecture is externally stateless in that no durable or persistent system state is stored outside of transient, register-local storage required solely to compute state transitions. Internal registers, buffers, or line memories may temporarily retain prior state values for computation, but no persistent or addressable memory state is maintained beyond immediate use.

Data Atoms and State Representation

Incoming data is segmented into discrete units referred to as data atoms. Data atoms may represent spatial, temporal, spatiotemporal, symbolic, logical, numerical, or abstract computational groupings. Atoms may be of arbitrary dimensionality, resolution, or bit-width and may correspond to pixels, voxels, signals, packets, tokens, instructions, feature vectors, or other structured or unstructured data.

Atoms are encoded into fixed-width or variable-width registers using deterministic ordering, mapping, or lookup mechanisms that preserve locality and structural meaning.

Delta Computation and State Evolution

Let P_t represent a current atom and P_{t-1} represent a prior atom stored in a register-local buffer. A delta Δ is computed as a function of P_t and P_{t-1}. In one embodiment, the delta is computed using a bitwise Exclusive-OR operation:

$$\Delta = P_t \oplus P_{t-1}$$

In alternative embodiments, delta computation may be performed using other reversible or irreversible operations, including XNOR, bitwise subtraction, bitwise addition modulo 2^n, masked comparison, arithmetic differencing, logical operations, algebraic transforms, or combinations thereof.

Register-Local Execution Core

The execution core comprises register-local storage, combinational or sequential logic for delta computation, and optional comparison logic. Deltas may be generated in deterministic time, including embodiments achieving single-cycle latency. External memory access is not required during normal operation.

Motif Encoding and Compression

Recurring delta patterns may be mapped to a vocabulary of motifs represented as indices, symbols, or tokens. Output transcripts record motif identifiers and positional or temporal context. Implicit zero-deltas may be omitted and deterministically inferred by a decoder.

Decoder and Reconstruction Symmetry

A complementary decoder applies received deltas to a synchronized prior state using an inverse or corresponding operation, enabling deterministic and lossless reconstruction. Encoder and decoder may operate asynchronously while remaining step-synchronized.

Parallel, Vectorized, and Distributed Embodiments

Multiple ATOMIK cores may operate concurrently on independent or interleaved data streams. Delta computation may be vectorized, pipelined, or parallelized. Distributed embodiments may partition atom streams across nodes while preserving deterministic reconstruction.

Hardware and Software Implementations

The invention may be implemented entirely in hardware, entirely in software executing on general-purpose processors, GPUs, or accelerators, or in hybrid hardware–software configurations. Software embodiments may emulate register-local behavior using caches, scratchpads, or constrained memory regions.

Security and Ephemeral Data Properties

Because no persistent external state is retained, intercepted delta streams lack reconstructible meaning without historical context. This property enhances security in embedded, edge, and physically exposed systems.

Higher-Order and Predictive Deltas

In some embodiments, multiple prior states (e.g., $P_{t-1}, P_{t-2}, \dots$) may be retained to support higher-order deltas, predictive coding, motion compensation, or trend-based state evolution.

Example HDL Implementation (Non-Limiting)

In one non-limiting embodiment, the architecture may be implemented in a hardware description language using an address-indexed local state memory. Upon receipt of a valid input atom, the prior state is retrieved, a delta is computed, the stored state is updated, and a zero-delta flag may be asserted.

```
module atomik_core (
    input wire clk,
    input wire rst_n,
    input wire valid_in,
    input wire [9:0] addr,
    input wire [3:0] pattern_in,

    output reg [3:0] delta_out,
    output reg is_zero,
    output reg valid_out
);
    reg [3:0] state_memory [0:624];
    reg [3:0] prev_state;

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            valid_out <= 0;
            delta_out <= 0;
            is_zero   <= 0;
        end else if (valid_in) begin
            prev_state = state_memory[addr];
            delta_out <= pattern_in ^ prev_state;
            state_memory[addr] <= pattern_in;
            is_zero   <= (pattern_in ^ prev_state) == 0;
            valid_out <= 1;
        end else begin
            valid_out <= 0;
        end
    end
endmodule
```

This example is illustrative only. Atom width, memory depth, addressing schemes, and delta operators may vary without departing from the invention.

Alternative Applications

The architecture may be applied to signal processing, video encoding, sensor fusion, secure communications, distributed systems, quantum control interfaces, real-time analytics, artificial intelligence preprocessing, and other domains requiring deterministic, low-latency computation.

Abstract

An externally stateless computing architecture that encodes computation as deterministic state evolution using register-local delta computation. Successive data atoms are compared using bitwise or algebraic operations to generate a transient delta stream representing system evolution without reliance on persistent external memory. The architecture reduces memory bandwidth requirements, improves energy efficiency, enhances security, and enables deterministic low-latency processing across hardware and software implementations.